



Ancient Egyptians were known to use probes, saws, forceps, scalpels and scissor, as well as clamps and sutures in their surgeries.



Sushruta, an ancient Indian physicians, is known as the father of surgery, known to have performed plastic surgery and rhinoplasty in Ancient India.

of this work, *On Surgery and Instruments*, received a lot of popularity and became the standard text for the next 500 years. It was also the first illustrated surgical guide. He is also known to have pioneered tracheotomy and neurosurgery.

Surgery underwent a transformation in the Middle Ages, during the 13th century, in the sense that a new type of craft emerged. Travelling Barber-surgeons would cut hair as other barbers, but they would also perform tooth extractions, bloodletting, bone settings, amputations and other small surgical operations. While physicians had started to go to medical school, surgery was still mostly learnt through apprenticeship and observation, not unlike the job of a blacksmith. Also, around this time, women started filling the role of the barber-surgeon as well, mostly because they were not allowed to become physicians. This continued till the 1700s when surgical training moved to universities, from which women were banned.

Two figures in the 16th century brought about major changes to the world of surgery. Andreas Vesalius, a professor in Padua around the time, broke the misconceptions about human anatomy which were based on animal dissection. His dissections on human cadavers led to the groundbreaking *De Humani Corporis Fabrica Libri Septem*, a text on anatomy that would stand the test of time for many years to come. He also advocated that every surgeon should get practical experience in dissection.

The second name is that of Ambrose Paré. The prevailing method of closing wounds was burning them shut (cauterizing, through boiling oil). Paré developed an emollient of egg yolk, rose oil, and turpentine for gunshot wounds, and also advocated for ligating (tying off) of blood vessels after amputation.

Before moving on to modern times, it is imperative to talk about John Hunter, the Scottish surgical scientist in the 1700s. Bringing an empirical approach to surgery, Hunter believed in observing and documenting everything by himself, without relying on third-party testimony. His work advanced our understanding of the venereal diseases and also advocated the study of pathology, infections and inflammations.

Modern (19th century)

The availability of better pain relief transformed surgery in the 19th century. By the 1840s, ether was being used by pioneers like Crawford Lang, and chloroform by John Snow,

physician to Queen Victoria. In addition to pain relief, it allowed physicians to conduct longer surgeries and perform more intricate operations.

The number of operations being undertaken started to grow, and with it came another problem - our lack of understanding regarding infections. Post-operative infections started to become quite common. The first step in curbing this came from Hungarian doctor Ignaz Semmelweis. He observed that particular students who went from the dissection room to the maternity ward were causing higher rates of maternal deaths as compared to midwives. Even under ridicule, he imposed compulsory handwashing and this led to a reduction in deaths in his ward. However, his advice was still ignored by the Royal Society.

The practice of using antiseptics wasn't accepted until Joseph Lister, based on the works of Louis Pasteur, carried out tests with carbolic acid on his instruments in the 1860s. The incidence of gangrene, his topic of study, immediately reduced and his paper, *Antiseptic Principle of the Practice of Surgery* saw widespread acclaim and set up practices that have prevailed.

Lister decided to take things one step further and instructed surgeons under him to use sterilised rubber gloves, wash their hands in 5 per cent carbolic acid solution, and use steam sterilised equipment. The empirical approach of Hunter, the discovery of modern anaesthesia, and Lister's sterilisation practices set the foundation for the invasive surgery we see today.

The present day (20th Century to now)

While diagnosis improved with the advent of X-ray in the late 19th century, the 20th century saw surgery become even safer and highly specialised. During the 1940s, the development of blood transfusion reduced death from blood loss, and penicillin greatly improved postoperative recovery. Over the years, the complexity of surgery significantly increased and saw aspiring surgeons study particular aspects of surgery for specialised qualifications.

The 1960s saw the first heart transplant and a heart bypass. Robots began assisting surgeons back in the 1980s, which also saw the first artificial heart used. Over the last 20 years, stem cell therapy, remote surgery, the first face transplant, multiple limb transplants, and many other achievements have completely redefined the pace of surgical advancements. 📺



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Robotic surgery and remote surgery are in practice today around the world and are helping save lives.